

Analysis General Exam August 2009

Closed book, closed notes. Please pledge. In each problem, justify all assertions, show calculations, and identify those theorems that you invoke in your arguments.

Problem 1

Suppose that f is a real-valued function on the real line $\mathfrak{R}$ with f absolutely continuous in the sense that it satisfies the fundamental theorem of calculus,

$$f(b) - f(a) = \int_a^b \frac{df(x)}{dx} dx$$

for $a, b \in \mathfrak{R}$, and for df/dx an $L^1(\mathfrak{R})$ -function, i.e., integrable over $\mathfrak{R}$.

- a) Use the definition of the Lebesgue integral and basic theorems of real analysis to show that given $\epsilon > 0$, there exists a simple function g of compact support so that $\|g - df/dx\| < \epsilon$, with $\|\cdot\|$ the L^1 norm.
- b) Prove that f is uniformly continuous and that $\lim_{x \rightarrow \infty} f(x)$ exists.

Problem 2

Let X be a measure space with measure μ and let $\{f_n\}$ be a sequence of real-valued measurable functions on X and f another measurable function so that

$$\mu \left\{ x \in X : |f(x) - f_n(x)| \geq \frac{1}{2^n} \right\} < \frac{1}{2^n}$$

- a) Show that the sequence $\{f_n\}$ converges to f pointwise, a.e.
- b) Suppose in addition that the functions of the above sequence are uniformly integrable in the sense that

$$\sup_n \int |f_n| d\mu \leq c$$

for some finite constant c . Show that f is integrable.

Problem 3

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ be the unit disc in the complex plane. For f analytic in $\mathbb{D}$ with power series

$$f(z) = \sum_{n \geq 0} a_n z^n$$

let

$$\|f\|_2 = \left(\sum_{n \geq 0} |a_n|^2 \right)^{1/2}$$

Let $\mathcal{F}$ be the collection of such analytic functions with finite $\|\cdot\|_2$ -norm.

a) Show that

$$\frac{1}{2\pi} \lim_{r \uparrow 1} \int_{\{z: |z|=r\}} |f(z)|^2 |dz| = \|f\|_2^2$$

($|dz|$ is differential arclength). Hint: Use polar coordinates.

b) Show that if $\{f_k\}$ is a Cauchy sequence in $\mathcal{F}$ with respect to the norm $\|\cdot\|_2$ defined above, then $\{f_k(z)\}$ converges uniformly for $z \in \mathcal{K}$, $\mathcal{K}$ any compact set contained in $\mathbb{D}$, and that the limiting function is analytic in $\mathbb{D}$.

Problem 4

Suppose that $P(z)$ is a polynomial of degree $n \geq 2$ with n distinct zeros $z_1, z_2, \dots, z_n$. Show that

$$\sum_{j=1}^n \frac{1}{P'(z_j)} = 0$$

Problem 5

Suppose that f is analytic in a open set containing the upper half plane $\{z : \operatorname{Im} z \geq 0\}$, and suppose further that for some $M, a > 0$, $|f(z)| \leq M/|z|^a$ for $|z|$ large. Show that for any z with $\operatorname{Im} z > 0$,

$$f(z) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{f(x)}{x - z} dx$$

Problem 6

True/False: Either prove that the statement is correct, or give a counterexample.

- (a) If f is analytic in the entire complex plane $\mathbb{C}$ and $\operatorname{Im} f \leq 0$ for all $z \in \mathbb{C}$, then f is constant.
- (b) If f is analytic in the entire complex plane and bounded on the real axis, then f is constant.
- (c) If f is analytic in the entire complex plane and $f(x+1) = f(x)$ for every real number x , then $f(z+1) = f(z)$ for every $z \in \mathbb{C}$.
- (d) If f is analytic in the entire complex plane with $|f(z)| \geq |z|$ for all $z \in \mathbb{C}$ and $f(1) = 1$, then $f(z) = z$ for all $z \in \mathbb{C}$.

Problem 7

Suppose that f is an analytic map of the unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ into itself that is not the identity function. Show that f can have at most one fixed point in $\mathbb{D}$.